# Aaron Zhou

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## Experience

**Software Engineer Intern** *May 2026 – Aug 2026*

*Ideogram* *Toronto, ON*

- **Migrated** the entire production ML inference fleet from TPU to GPU infrastructure, **freeing 2,000+ TPUs for model training** while reworking model serving and deployment pipelines for GPU-native execution
- Built and shipped **Ideogram V4 Remix**, a production image-to-image generation pipeline spanning frontend, APIs, model routing, and captioning
- **Reduced ML serving cold-start time by 13.5%** (7+ minutes) through model compilation, caching, and warm-up optimizations across **GPU/TPU** infrastructure
- **Optimized latency across 10+ model architectures**, **halving LLM prompt generation latency** (10s → 5s) with **SGLang** and **cutting diffusion serving latency by 40%** via CPU/GPU pipeline parallelization
- **Distilled** 2M+ user-generated images into a **15K curated dataset** for style-transfer LoRA fine-tuning, building data processing and quality-filtering pipelines for image-to-image training

**Computer Vision Software Engineer Intern** *May 2025 – Dec 2025*

*Martinrea International - Advanced Manufacturing Team (StationVision)* *Toronto, ON*

- Secured **€1.2 million in a production contract from Tesla** by designing and implementing a **real-time computer vision system from the ground-up** in Python, using YOLO with camera and PLC integration
- Engineered the system to detect a wide range of manufacturing defects supporting **21** distinct part variants across seven production cells and three hardware/software architectures, using **object detection and classification** models to ensure complete quality control on **300,000+ parts and counting**
- Achieved an **8× runtime speedup** by parallelizing across NVIDIA Jetson cores using Python multiprocessing
- Delivered an urgent proof-of-concept computer vision system by performing data preprocessing and training three models, **de-risking deployment and earning plant approval** for our Tesla vision system implementation

**Computer Vision Software Engineer Intern** *Jan 2025 – May 2025*

*Martinrea International - Advanced Manufacturing Team (StationVision)* *Toronto, ON*

- Achieved $**84,000 in annual savings** by taking over and enhancing an existing weld quality detection system in 2 production cells, **improving detection accuracy from 80% to >99.9%** while fully eliminating false negatives
- Designed a **self-recovery mechanism** with Python and systemd, protecting the weld system from power outages and hardware crashes, reducing failures by **50%+**
- Developed a **synthetic data augmentation script** to expand training data for **30+** production models with previously limited data, enhancing model performance by up to **10%**
- Presented our vision software innovations to **key investors and executives** (CEO of Martinrea, CEO of Equisphere), media (CBC, CNN), and politicians (Prime Minister Mark Carney, Mayor Steven Del Duca)

**Full-Stack Software Developer Intern** *May 2024 – Aug 2024*

*RBC Capital Markets - Global Investment Banking* *Toronto, ON*

- Designed a **caching layer** in order to efficiently query an **SQL** database of **over 10,000+ deals and transactions** and integrated **AngularJS** front-end with **C#** backend
- Implemented a **React** interactive table visualization used by energy analysts at **100+** client companies

**Software Developer Intern** *May 2023 – Aug 2023, Jan 2024 – Apr 2024*

*Canadian Imperial Bank of Commerce (CIBC) - DevOps* *Toronto, ON*

- Automated workflows of request tickets for **Jira, Confluence, GitHub, Artifactory**, and **Jenkins**
- Migrated project tracking request forms to Jira Service Desk, enabling search and sort for **80,000+ requests**
- Created a Github Action integration using JavaScript and GraphQL to query the Github API for user permissions, saving the auditing team 1000+ hours annually

## Education

**University of Waterloo** *Sep 2022 – Apr 2027*

*Honors Bachelor of Computer Science (Co-op)* *Waterloo, ON*

## Projects

**MeepMeep — *Kotlin, Supabase***

- Built an Android bird-logging app with **Kotlin Composables**, featured as a top project on the course website.
- Integrated **speech-to-text recognition** for complex bird names and a **customizable interactive map** using OpenStreetMaps, allowing for real-time location tracking

## Technical Skills

**Languages**: Python, C, C++, C#, Bash, SQL, JavaScript, HTML/CSS, Kotlin, Java, R
**Frameworks**: PyTorch, Jetpack Compose, Android SDK, React, AngularJS, Django, GraphQL, Material UI, Swagger
**Tools & Libraries**: Linux, Git, YOLO, Docker, Kubernetes, vLLM, SGLang, Android Studio, Figma, NumPy, Ultralytics, OpenCV, OpenAI